Self-Diagnosis Software

Laser Diagnosis Manual

Applicable model

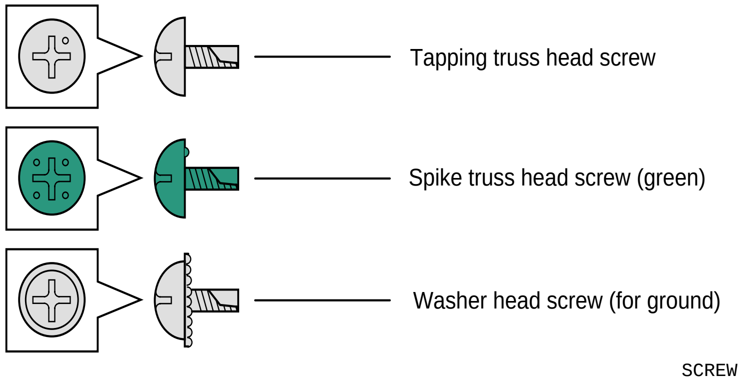
- QSS-30 series

Issued in April, 2006

Notes to service personnel

Be sure to read this manual carefully to gain a thorough understanding of the correct procedures before servicing the system.

The printer processor uses both tapping truss head screws and spike truss head screws. When attaching the screws once removed, make sure they are on their original positions. These screws are used for the place where grounding is required.



- It is prohibited to show, provide, lend or transfer this manual to the others except the service personnel.
- The contents of this manual are subject to change without notice.
- Illustrations in this manual may vary depending on the model or manufacturing lot.

Export Control

This Product shall be exported pursuant to the Japan Foreign Exchange and Foreign Trade Control Law, and Export Control of Administration on Regulations of the U. S. Bureau of Industry and Security, and relevant laws and regulations. If Buyer re-exports, directly or indirectly, the product to any country, the rules or regulations immediately preceding and relevant laws and regulations of the countries concerned shall be applied.

Explanation of manual

About the chapters

Table of Contents of Laser Diagnosis Manual

The Laser Diagnosis Manual is designed to help solve problems related to the laser using the Self-Diagnosis Software.

- 1. Cautions for work
  Contains information on how to achieve safety in service operations. Be sure to read precautions thoroughly and carefully.
- 2. Outline of Self-Diagnosis Software
  Describes the outline of the Self-Diagnosis Software.
- 3. How to use Diagnosis Software
  Explains how to start up the Self-Diagnosis Software, the screens and the diagnosis procedures.
- 4. Examples of diagnoses
  Gives examples of the diagnoses and explains the situations in which this software can and cannot be used.

Marks and symbols used in this manual

Definitions of the marks and symbols used in this manual are as follows:

⚠ This is called the alert symbol mark. Text following this mark contains particularly important information concerning safety. Be sure to heed this information. This mark is used in conjunction with the words **DANGER**, **WARNING** and **CAUTION**, according to the extent of influence (injury) on persons or damage to physical property.
IMPORTANT The Important symbol indicates operations or procedures requiring caution, instructions and supplementary explanations that need to be followed.
☞ The pointing finger symbol indicates the manual or section where you can find additional information.
NOTE The Note symbol indicates functions or instructions which are convenient if you know.

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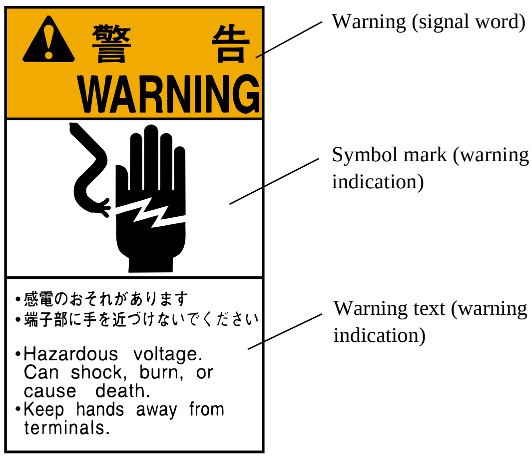
1. Cautions for work

1.1 Description of warning (signal words)

- Signal words identify the level of injuries that can potentially occur.
- The signal words used in this manual and found on labels, **DANGER**, **WARNING** and **CAUTION**, are assigned according to the level of potential risk.
- Warning labels are located at or near the part of the system that pose the indicated danger. If ignored, death or serious injury occurs, or the system breaks down. Be sure to follow the indications in the manuals and on the warning labels.
- The warnings include a signal word, the type and extent of the danger, and information to avoid danger.
- Carefully read and follow the warnings included in this manual and on the warning labels before operating the system.

⚠ DANGER This indicates situations that if not immediately avoided could result in serious injury or death.
⚠ WARNING This indicates situations that if not avoided could result in serious injury or death.
⚠ CAUTION This indicates situations that if not avoided could result in non-life threatening injury. It is also used to indicate situations which may cause damage to physical property.

Example of warning label



SIGNALWORD

1.2 For safe operation

1.2.1 General precautions

WARNING

- Prior to any part replacement or mechanical adjustment, be sure the main power supply is turned off.
- Since the work which uses key operations cannot turn off the circuit breaker, mechanical operation check during it requires particular attention.

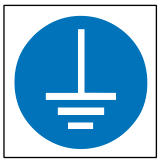
IMPORTANT

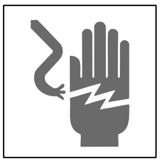
- Ground wires (green and yellow) are connected to the covers and units of the system.
  For reassembly, be sure to connect the ground wires as they were.
- Be sure to perform an operation check after replacing or adjusting any parts (or units).

1.2.2 Precautions against electric shock

WARNING

- If any case you have to take care of wiring for the power such as moving the system, ask a qualified professional electrician for work. Do not forget to ground the system.
- Pay attention to avoid shocks when performing troubleshooting, wiring checking, or voltage/current measurement.
- When replacing a fuse or PCB, be sure to turn off the circuit breaker and the main power supply. Wait for 10 seconds or more before replacement.





1.3 Countermeasure for static electricity when replacing and maintaining the electronic parts

If an electrically charged human body touches electronic parts such as PCBs, it may adversely affect the electronic parts. When handling the electronic parts, be sure to use static-dissipative tools as below to prevent the components on the PCB from being damaged due to static electricity.
Also use the static-dissipative tools for maintenance of the digital units or engines.

Static-dissipative tool

Description	Remarks
Portable Static-Dissipative Field Service Kit	Use this kit when replacing or installing/removing the electronic parts from the machine. This kit consists of four items of Static-Dissipative Work Mat, Wrist Strap, Ground Cord, and Alligator Clips.
Static-Dissipative conductive gloves	Use this to prevent that sebum on your hand adheres when you touch a PCB.
Wrist strap	Use this when checking the electronic parts.

CAUTION

- When using the static-dissipative tool, be sure to turn off the circuit breaker of the unit and the main power supply, and wait 10 seconds or more to carry out operation.

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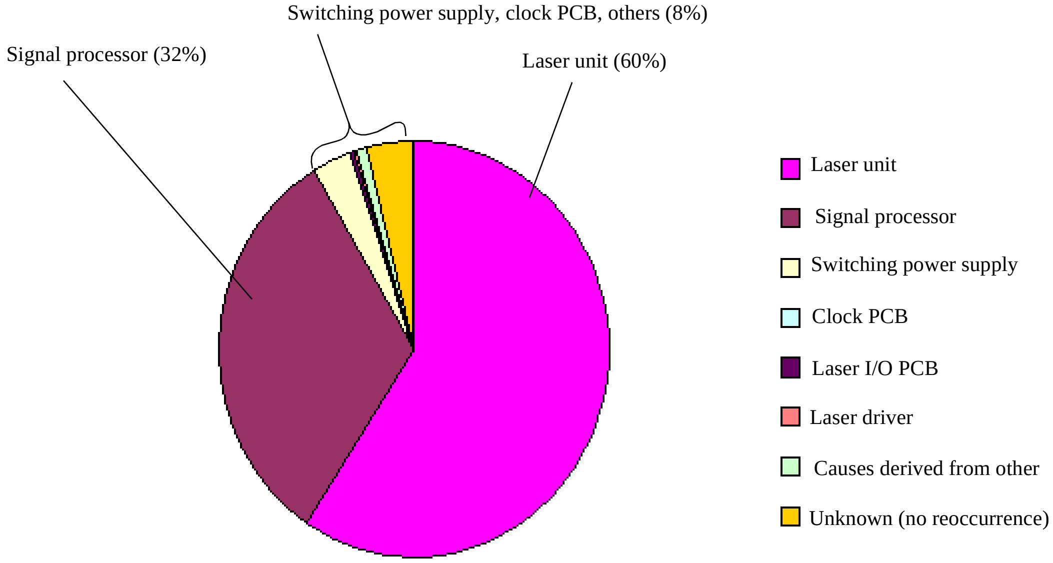
2. Outline of Self-Diagnosis Software

2.1 Outline of Self-Diagnosis Software

2.1.1 Before using

Analyzing the past data of the QSS-30 series shows that 60% of the problems related to lasers are failures in the laser unit (such as AOM diffuseness, LD deterioration and optical axis fluctuation of R laser) and 32% of them are malfunctions of the signal processor (AOM driver). These troubles alone account for more than 90% of the whole.
For this reason, in troubleshooting, most of the problems can be solved if it is possible to sort them into either of these two causes. However, both may cause phenomenon of laser light intensity lowering so that some problems may be difficult to diagnose.
In addition, it often happens that the phenomenon does not reoccur during the service personnel's visit. This makes the situation more difficult.
Therefore, as a useful tool to judge the normality of the laser unit and signal processor, we have created the Laser Self-Diagnosis Software.

Cause of the problems related to the laser with the QSS-30 series (analysis of data from the past 1.5 years)



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2.1.2 When to use

Although various symptoms appear with the troubles related to laser, this Self-Diagnosis Software cannot be used for all of those symptoms.
The following are the symptoms for which the Self-Diagnosis Software can and cannot be used.
For details, refer to ☞ Examples of diagnoses.

Symptoms for which the Self-Diagnosis Software can be used

- If **No. 6073: Synchronous Sensor error.** occurs
- If **No. 6105: B Laser light source status error.** or **No. 6106: G laser light source status error.** occurs

Symptoms for which the Self-Diagnosis Software cannot be used

- If the print color fluctuates (light intensity fluctuation)
- If abnormal print with noise is made
- For errors at the laser temperature adjustment

IMPORTANT

- This Self-Diagnosis Software cannot be used when the QSS software is running. Be sure to finish the QSS software before using.
- The Self-Diagnosis Software has the function to perform voluntarily the synchronous detection which is same as print operation.
  To use this function, it is necessary to make the laser unit under the same condition as that to print. For this reason, it is required to finish the temperature adjustment in the laser unit.

If replacing the laser unit and checking with the Self-Diagnosis Software, a certain waiting time is necessary until the laser temperature adjustment is finished.

2.1.3 Compatible model types and versions

Compatible model	Supported version
QSS-3001, 3011*1, 3000, 3010	Ver. D or later
QSS-3020S, 3021S	Ver. A or later

*1. QSS-3011DLS is not supported.

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3. How to use Diagnosis Software

3.1 Starting/quitting Self-Diagnosis Software

3.1.1 Starting Self-Diagnosis Software

IMPORTANT

- To start the self-diagnosis software when it has already been upgraded to the self-diagnosis version, go to ☞4. (for example, to diagnose again after changing the signal processor)

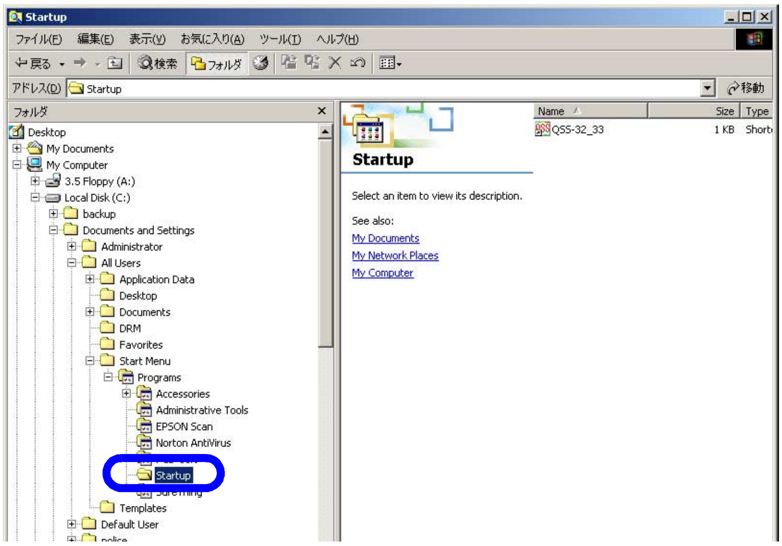
1. Save the self-diagnosis software to a readable and writable storage media.

NOTE
- The data size of this self-diagnosis program is about 400 KB.
- You can use the self-diagnosis software data file saved in a new folder created in the drive C:\QSS.

2. If the QSS software is running, quit it. Display the Windows screen.

NOTE
- The self-diagnosis software and QSS software cannot be used at the same time.

3. Move the QSS software (QSS-##) from Startup folder to desktop.



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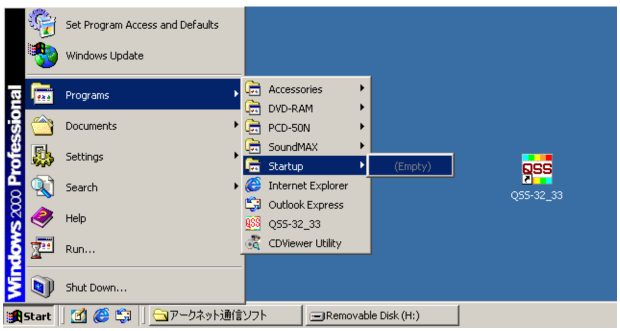
NOTE
- QSS software is moved from Startup to Desktop so that the QSS software does not start automatically after restarting the system.
- You can also access the Windows program menu from **Start**→**Programs**→**Startup**, and drag the QSS-## file to the desktop from there.



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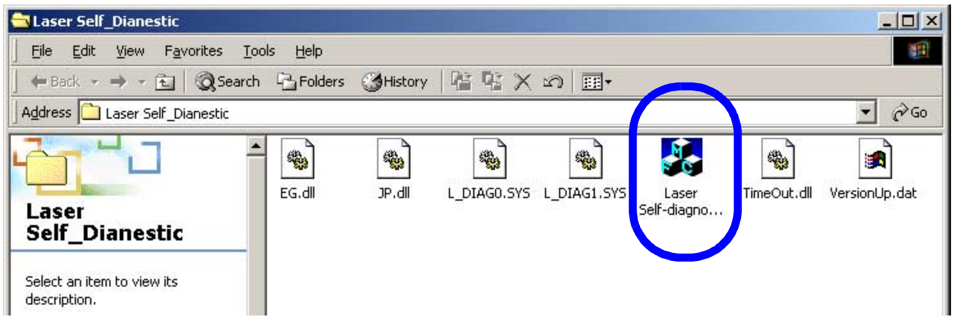
4. Insert the storage media with the self-diagnosis software into the drive, and double-click **Laser Self-Diagnostic.exe**.



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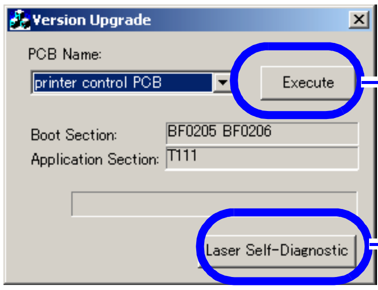
5. The **Version Upgrade** display appears. Click **Execute** or **Laser Self-Diagnosis**.



If the software has not been upgraded to the self-diagnosis version

If the software has been upgraded to the self-diagnosis version

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NOTE

- After the software is started, PCB name to be upgraded is shown on the **PCB Name** list.
  You do not have to select the PCB name even though the pulldown menu is displayed.
- If the software has already been upgraded to the self-diagnosis version, click **Laser Self-Diagnostic** to go to step ☞ 7.

6. After the message **Would you like to perform version upgrade for** [###### **PCB**]**?** appears, click **OK**. PCB upgrading starts automatically.

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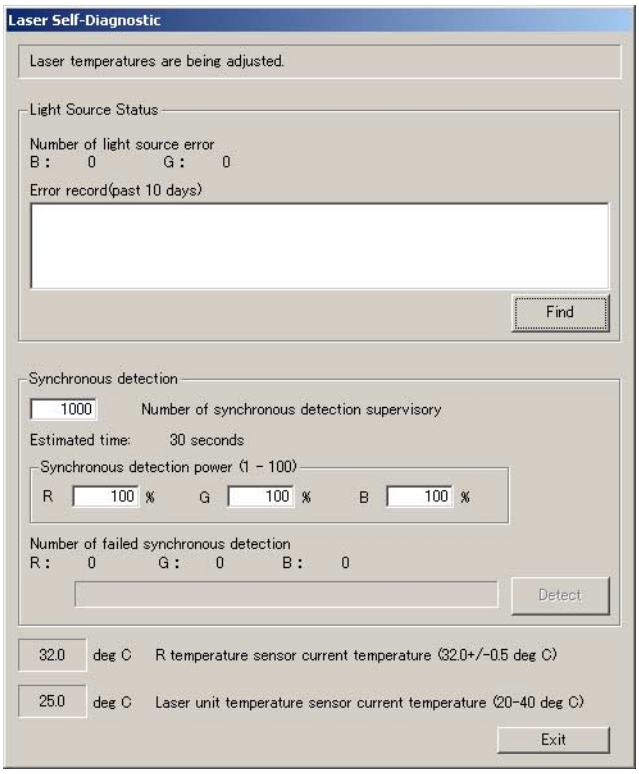
7. The **Laser Self-Diagnostic** display appears automatically.



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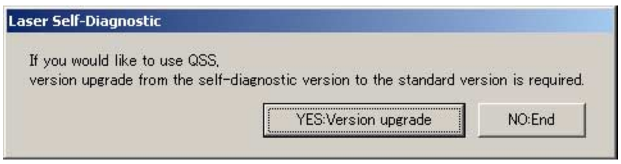
8. This completes starting up the software.

3.1.2 Quitting Self-Diagnosis Software

1. Click **Exit** on the **Laser Self-Diagnostic** display.

2. Click the Close button at the top right of the **Version Upgrade** display.

3. After the message **If you would like to use QSS, version upgrade from the self-diagnostic version to the standard version is required.** appears, click **YES:Version upgrade**.



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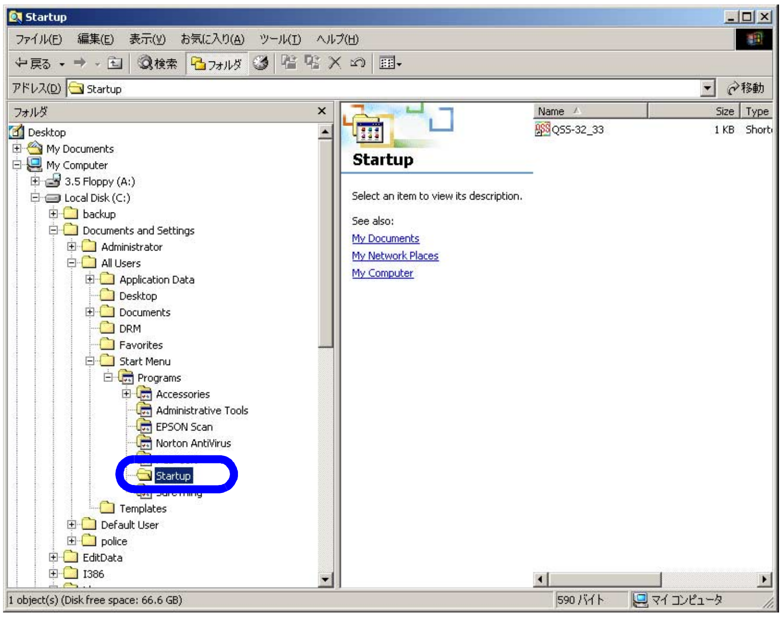
NOTE

- Clicking **YES:Version upgrade** changes the program to the standard version, then you can use the QSS software.
- To turn off the power supply in order to shift the signal processor, click **NO:End** to quit the program while remaining in the self-diagnosis version.

4. Move the QSS software (QSS-##) back to Startup folder.

Move the **QSS-##** file from Desktop to **Startup**.



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NOTE

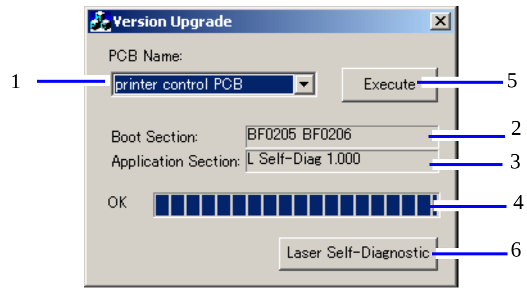
- You can also move the **QSS-##** file from Desktop to **Startup** by dragging and dropping the **QSS-##** file via **Start→Programs→Startup**.

5. This completes quitting the software.

3. How to use Diagnosis Software

3.2 Explanation of Self-Diagnosis software display

3.2.1 Version Upgrade display



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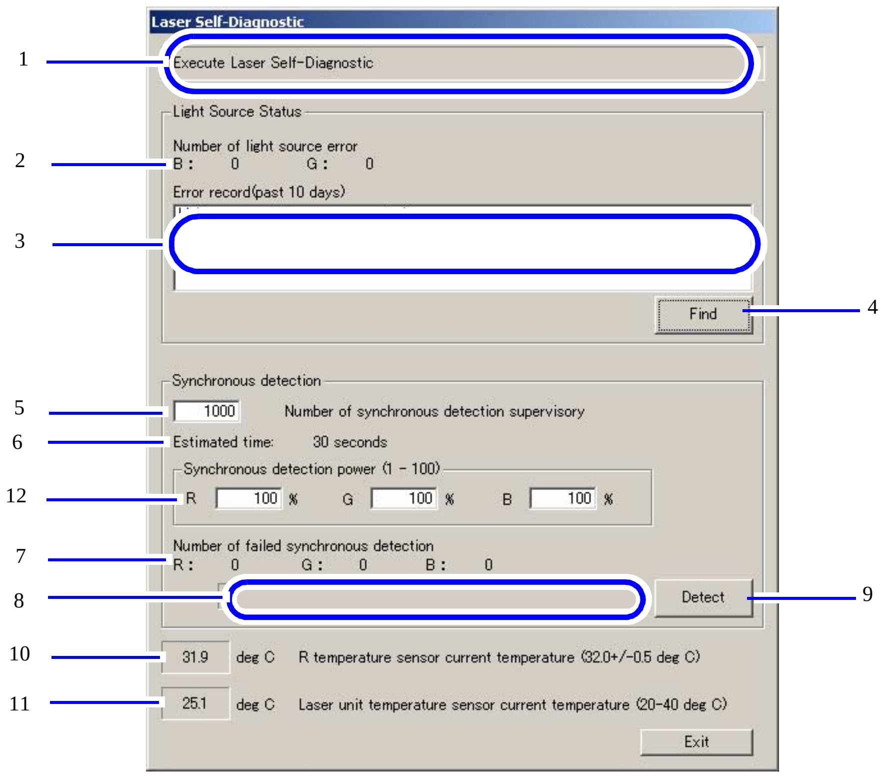
The numbers in the table below correspond to each number in the screen above.

No.	Item	Explanation
1	PCB Name	Shows the PCB to be upgraded.
2	Boot Section*1	Shows the version of PCB Boot section.
3	Application Section*1	Shows the version of PCB application section.
4	Upgrading status	Shows the upgrading status by a progress bar.
5	**Execute** button	Starts upgrading the selected PCB.
6	**Laser Self-Diagnostic** button	Displays the Laser Self-Diagnostic display. (This button is activated only when the PCB is in self-diagnostic version.)

*1. Not displayed when the QSS is running.

3.2.2 Laser Self-Diagnostic display



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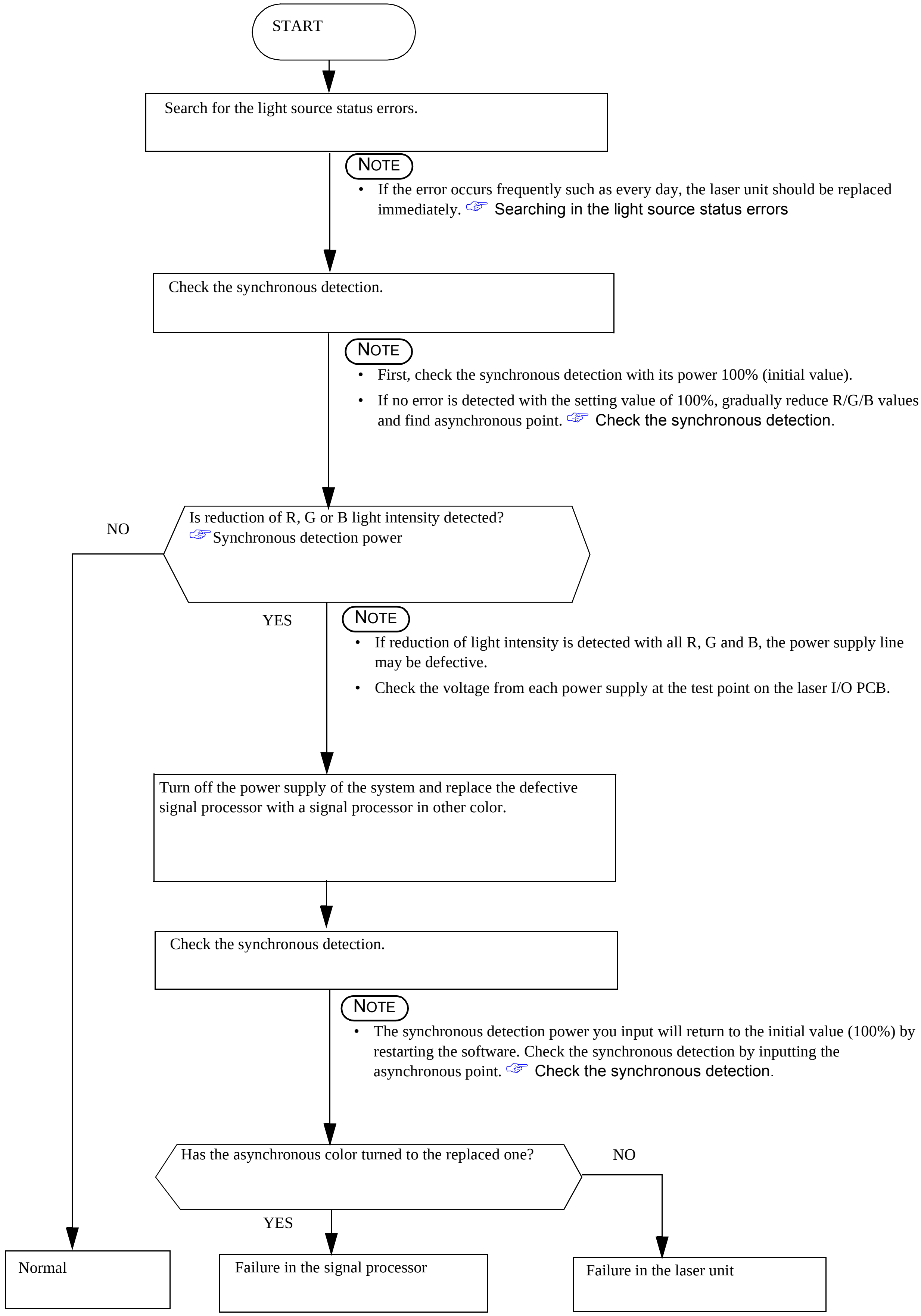
The numbers in the table below correspond to each number in the screen above.

No.	Item	Explanation
1	Laser Self-Diagnosis software status	Shows the status of the Laser Self-Diagnosis software.
2	Number of light source error	Searches for the light source status errors and shows each number of occurrences of **B Laser light source status error.** and **G Laser light source status error.** • For details, refer to ☞ Searching in the light source status errors.
3	Error record	Searches for the light source status errors and shows each detail of occurrences of **B Laser light source status error.** and **G Laser light source status error.**
4	**Find** button	Searches the number of the light source status error occurrences in the QSS error record (BkData0100.dat) within the last 10 days.
5	Number of synchronous detection supervisory	Configures how many times synchronous detection observation is performed for each of R, G and B. • For details, refer to ☞ Number of synchronous detection supervisory.
6	Estimated time	The estimated time (sec.) for synchronous detection observation is automatically calculated by entering the number of synchronous detection observation.
7	Number of failed synchronous detection	Shows the number of failed synchronous detection for each R, G and B. The number of failed synchronous detection will be updated when completing the synchronous detection.
8	Synchronous detection progress bar	Shows the time passage of the estimated time.
9	**Detect** button	Starts the R, G and B synchronous detection. While detecting, the **Stop** button appears. Clicking this button stops the synchronous detection. **IMPORTANT** • Synchronous detection cannot be performed while adjusting the laser temperature.
10	R temperature sensor current temperature	Shows current temperature detected by R temperature sensor.
11	Laser unit temperature sensor current temperature	Shows current temperature detected by the laser unit temperature sensor.
12	Synchronous detection power	Configures each laser power for checking synchronous detection. Using this function, check how much leeway is available in the synchronous detection level. • For details, refer to ☞ Synchronous detection power.

3.3 Diagnosis flow chart

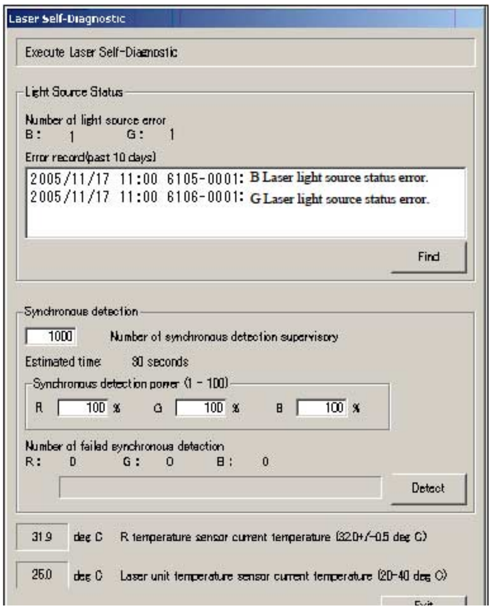


3.4 Details of diagnosis

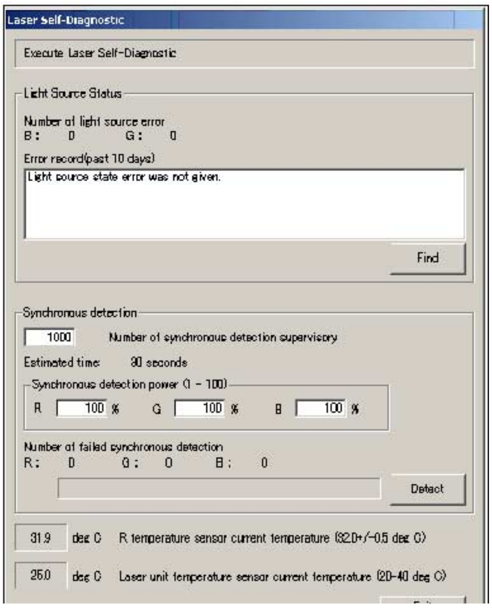
3.4.1 Searching in the light source status errors

Click **Find** to search for errors in the error record up to the last 10 days. If **No. 6105: B Laser light source status error.** or **No. 6106 G laser light source status error.** is found, the number of each error occurrence appears and the error record content is displayed on the list.

When errors occurred:



When no error occurred:



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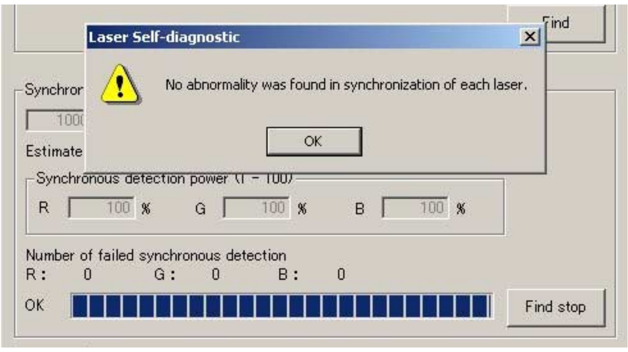
IMPORTANT

- The errors listed above mean that the LDs (laser diode) built in the B and G laser heads are deteriorated.
- If either of the errors above occur, past experience has shown that there is a high probability that the laser unit is the cause.
  If the error occurs every day although it used to occur at a few days intervals at first and comes to occur with more increased frequency so that it cannot be released, the laser unit must be replaced urgently.
- The progression of LD deterioration lowers the laser light intensity. By the asynchronous point explained in the following, also check the level of the laser light intensity lowering and use it as a judgment item for replacement timing.

3.4.2 Check the synchronous detection.

Based on the setting values of number of synchronous detection supervisory and synchronous detection power, the system observes the synchronous detection.
As the observation result, number of the failed synchronous detections is displayed separately for R/G/B with the message below.
If detection of R, G or B fails even once, it is judged as abnormal.

Message displayed if finished normally



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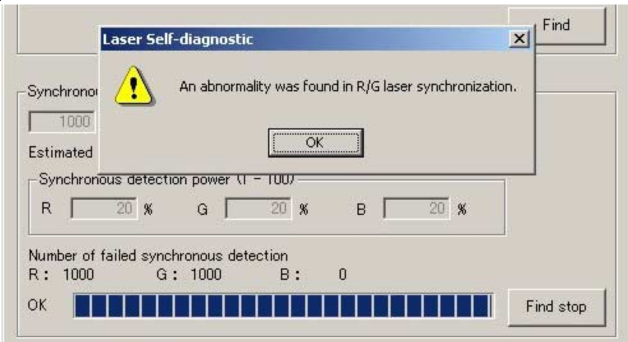
Message displayed if not finished normally

Example: When synchronous detection failed in R/G lasers



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● Number of synchronous detection supervisory

Enter how many times synchronous detection observation is performed.

- Input range: 1000 to 60000 (initial value: 1000)
- Time required: 30 seconds to 30 minutes

IMPORTANT

- Inputting larger value increases the number of synchronous detections. At the same time, the observation time becomes longer.
  For example, for cases with low occurrence frequency, change the value if you want to observe for longer time.
  When the symptom is occurring currently, 1000 times is sufficient for observation.

● Synchronous detection power

You can set the laser power (light intensity) for synchronous detection.

- Input range: 100 to 1 (initial value: 100)

First, enter 100 for checking.
If no abnormal condition is detected with the laser power of 100%, gradually reduce the R, G and B values to detect abnormal condition.

IMPORTANT

- With the usual synchronous detection such as that when printing, the detection is performed with the power of 100%. Decreasing this value reduces the light intensity, and no laser light is output with the lowest value, 1%.
  Thus, the point at which the synchronous detection cannot be performed (asynchronous point) must exist somewhere between 100 to 1%. Asynchronous point tends to become larger with less light intensity and smaller with more light intensity.
  For example, asynchronous point is at 100% for system with little laser light output and with high occurrence frequency of **No. 6073: Synchronous Sensor error.**, 70% for system with lower laser light and with low occurrence frequency of **No. 6073: Synchronous Sensor error.**, 30% for system with correct laser light intensity.

- As the standards of judgment with regard to laser unit or signal processor, use the following values.

Asynchronous point	Judgment
100 to 60%	Light intensity has deteriorated (the closer to 100%, the greater urgency)
59 to 40%	Light intensity tends to deteriorate (Even though the urgency is not high, enough attention should be paid.)
39 to 20%	Light intensity has not deteriorate (adequate light intensity)

NOTE

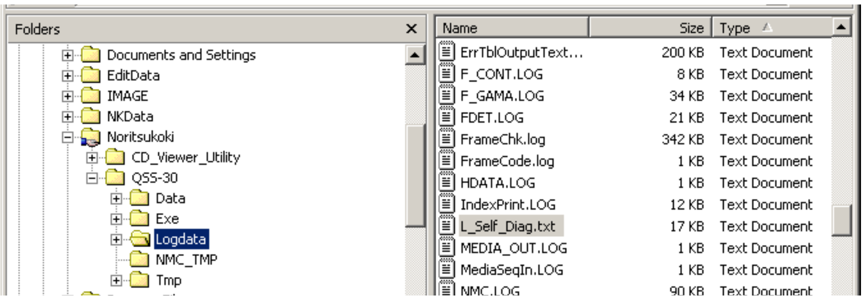
- Even if the asynchronous point is at less than 39%, this does not guarantee normal functioning.
  For details, refer to ☞ Examples of diagnoses.

3.5 Automatic saving of diagnosis result

3.5.1 Location for saving diagnosis data

This self-diagnosis software automatically writes the diagnosis data to the **L_Self_Diag_.txt** file in the **C:\Noritsukoki\QSS-##\Logdata** folder.

- Software upgrade information
- Result of checking the light source status
- Result of synchronous detection
- Error



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3.5.2 Information in the saved file

Following information is shown from the left with TAB intervals.

Operation date

- The start and end time of upgrading is shown.

PCB Name

- The name displayed at **PCB Name** to be upgraded is shown.

Boot

- The version of the boot section is shown.

Application

- The version of the application section is shown.

Example: Description in the saved file

```
//□□□□                 □□□                         Boot                                         Aplication
2006/01/27 09:30:47    printer control PCB    BF0205 BF0206         L Self-Diag 1.000
2006/01/27 09:32:17    □□□□□□□□□ [ B:   0□ G:   0□ ]
2006/01/27 09:37:51    □□□□□□□□[  1000□ ] □□□□□□□[ R:100 G:100 B:100 ] □□□□□□[ R:    0□ G:    0□ B:    0□ ]
2006/01/27 09:38:52    □□□□□□□□[  1000□ ] □□□□□□□[ R: 90 G: 90 B: 90 ] □□□□□□[ R:    0□ G:    0□ B:    0□ ]
2006/01/27 09:39:35    □□□□□□□□[  1000□ ] □□□□□□□[ R: 80 G: 80 B: 80 ] □□□□□□[ R:    0□ G:    0□ B:    0□ ]
2006/01/27 09:40:20    □□□□□□□□[  1000□ ] □□□□□□□[ R:  1 G: 80 B: 80 ] □□□□□□[ R: 1000□ G:    0□ B:    0□ ]
2006/01/27 09:41:05    □□□□□□□□[  1000□ ] □□□□□□□[ R: 50 G: 80 B: 80 ] □□□□□□[ R:    0□ G:    0□ B:    0□ ]
2006/01/27 09:41:44    □□□□□□□□[  1000□ ] □□□□□□□[ R: 30 G: 80 B: 80 ] □□□□□□[ R: 1000□ G:    0□ B:    0□ ]
2006/01/27 09:42:32    □□□□□□□□[  1000□ ] □□□□□□□[ R: 35 G: 80 B: 80 ] □□□□□□[ R: 1000□ G:    0□ B:    0□ ]
2006/01/27 09:43:15    □□□□□□□□[  1000□ ] □□□□□□□[ R: 40 G: 80 B: 80 ] □□□□□□[ R:    0□ G:    0□ B:    0□ ]
2006/01/27 09:43:53    □□□□□□□□[  1000□ ] □□□□□□□[ R: 40 G: 40 B: 80 ] □□□□□□[ R:    0□ G:    0□ B:    0□ ]
2006/01/27 09:44:31    □□□□□□□□[  1000□ ] □□□□□□□[ R: 40 G: 30 B: 80 ] □□□□□□[ R:    0□ G:    0□ B:    0□ ]
2006/01/27 09:45:10    □□□□□□□□[  1000□ ] □□□□□□□[ R: 40 G: 20 B: 80 ] □□□□□□[ R:    0□ G: 1000□ B:    0□ ]
2006/01/27 09:46:00    □□□□□□□□[  1000□ ] □□□□□□□[ R: 40 G: 30 B: 30 ] □□□□□□[ R:    0□ G:    0□ B:    0□ ]
2006/01/27 09:46:38    □□□□□□□□[  1000□ ] □□□□□□□[ R: 40 G: 30 B: 20 ] □□□□□□[ R:    0□ G:    0□ B: 1000□ ]
```

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4

4. Examples of diagnoses

4.1 Examples of diagnoses

The examples here are those diagnosed using the laser self-diagnosis software when analyzing defects.

NOTE

- The self-diagnosis software is used with the conditions below.
  - Number of synchronous detection supervisory: 1000
  - Start the check of synchronous detection with the power of R/G/B 100% and repeat changing each value and checking in order to find the asynchronous point.

In some cases, the laser self-diagnosis software is not effective.

When the self-diagnosis software is effective:

Reference	Symptom	Cause
☞Example (1)	The laser light source status error and synchronous sensor error appear.	Laser unit
☞Example (2)	Synchronous sensor error appears.	Laser unit
☞Example (3)	Synchronous sensor error appears.	Signal processor

When the self-diagnosis software is ineffective:

Reference	Symptom	Cause
☞Example (4)	The colors printed in the morning and afternoon differ by one or two keys.	Laser unit
☞Example (5)	The print color suddenly changes to red or cyan by two or three keys.	Signal processor

IMPORTANT

- Analysis of defects in cases such as (4) and (5) shows that the asynchronous point was generally at less than 39% intensity in laser units and signal processors with confirmed failure.
  These defects occurred before the light intensity degradation began and the symptoms are from phenomena other than laser asynchronicity, such as print color fluctuation.

4.1.1 Examples that can be checked with the self-diagnosis software

● Example (1)

Symptom

- Both **No. 6105: B Laser light source status error.** and **No. 6073: Synchronous Sensor error.** occur every morning.
  No. 6073: Synchronous Sensor error. occurs even in processing prints.

Corrective action

- The B laser driver and B signal processor were changed, but the error still appeared.
  Finally, this error was cleared by replacing the laser unit.

Analyzed result

- The laser unit: LD (laser diode) is deteriorated and deterioration of light intensity of B laser is detected.
- Signal processor: Normal
- B laser driver: Normal

Diagnosis result of Self-Diagnosis Software

- Deterioration of B laser light intensity is detected.
  (The following is the check result by installing the laser unit returned to the factory.)

Each laser	Asynchronous point	Remarks
R	35%	Normal
G	32%	Normal
B	76%	Abnormal

Example (2)

Symptom

- The error message **No. 6073-003: Synchronous Sensor error.** appears every morning.
  The error is cleared by resetting the power supply and the same error does not appear within a day.
  The error used to occur every few days. However, the frequency increased and now occurs every day.

Corrective action

- The error was cleared by replacing the B signal processor and laser unit.

Analyzed result

- Laser unit: AOM diffuseness is detected in B laser.
- Signal processor: Normal

Diagnosis result of Self-Diagnosis Software

- Deterioration of B laser light intensity is detected.
  (The following is the check result by installing the laser unit returned to the factory.)

Each laser	Asynchronous point	Remarks
R	35%	Normal
G	32%	Normal
B	64%	Abnormal

Example (3)

Symptom

- **No. 6073-0002: Synchronous Sensor error.** suddenly occurs and the error cannot be released.

Corrective action

- The error was cleared by replacing G laser signal processor.

Analyzed result

- Signal processor: Element in the circuit was broken and the output (electric power) was 0 mW.

Diagnosis result of Self-Diagnosis Software

- Deterioration of light intensity is detected.
  (The following is the check result by attaching the returned signal processor to G laser.)

Each laser	Asynchronous point	Remarks
R	32%	Normal
G	100%	Abnormal
B	25%	Normal

4.1.2 Examples that cannot be checked with the self-diagnosis software

Example (4)

Symptom

- The colors printed in the morning and afternoon differ by one or two keys.

Corrective action

- Replace the laser unit.

Analyzed result

- Laser unit: AOM diffuseness is detected in G laser.

Diagnosis result of Self-Diagnosis Software

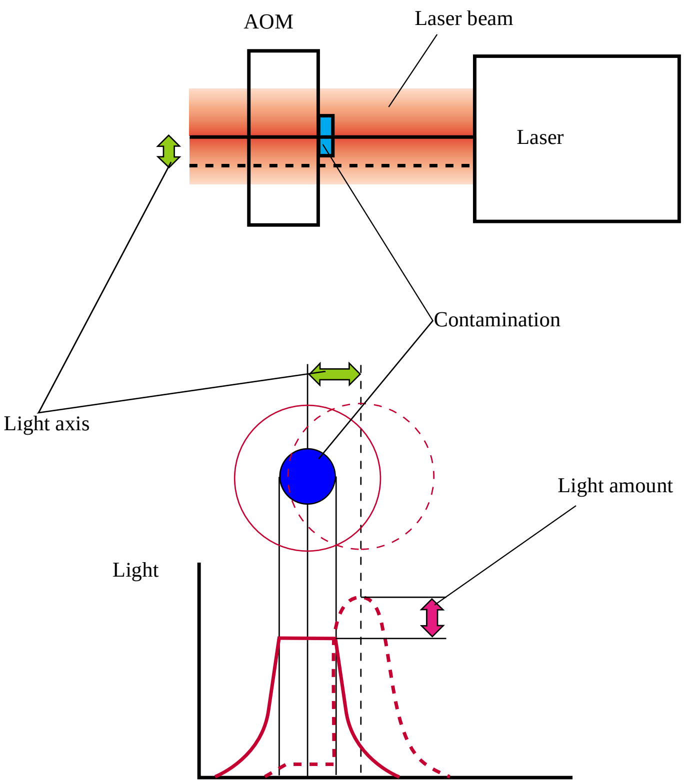
- Deterioration of light intensities is not detected in R, G and B lasers.

(The following is the check result by installing the laser unit returned to the factory.)

Each laser	Asynchronous point	Remarks
R	30%	Normal
G	25%	Normal
B	25%	Normal

NOTE

- If contamination attaches on the surface of AOM crystal, the light entered to AOM diffuses and causes defects. AOM diffusion mainly causes fogged prints and the error **No.6073: Synchronous Sensor error.** due to deterioration of light intensity. In addition, it may cause color change of print.
- The mechanism of print color fluctuation due to contamination attachment is as follows.



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The light axis of each R, G and B laser will slightly change depending on temperature alteration.
Usually, this change hardly affects the print color.
However, if any contamination attaches on AOM around the center of the laser beam (peak part with the highest light intensity), the peak part may be interrupted or appear because of minute variation of light axis.
Variations change the print colors greatly.
However, with this level, the influence on the synchronous sensor is limited and the asynchronous point may show the normal range.

● Example (5)

Symptom

- The print color suddenly changes to red or cyan by two or three keys.

Corrective action

- The error was cleared by replacing R laser signal processor.

Analyzed result

- Signal processor: Element in the circuit was broken and the signal processor output status was unstable.

Diagnosis result of Self-Diagnosis Software

- Deterioration of light intensity was not detected.

(The following is the check result by attaching the returned signal processor to R laser.)

Each laser	Asynchronous point	Remarks
R	30%	Normal
G	25%	Normal
B	25%	Normal

NOTE

- If the signal processor is broken, electrical power may not be output at all. ☞ Example (3)
  However, in many cases, the output may swing and be out of the normal range after some time.
  When diagnosing with normal output, light intensity degradation will not be detected. Also, if diagnosing when the output slightly falls from the normal range, the influence on synchronous sensor is limited and the asynchronous point may show the normal range.

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